MILITARY SPECIFICATION

STERILIZER, SURGICAL INSTRUMENT AND DRESSING, LABORATORY

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a lightweight, single-chambered jacketed, horizontal pressure sterilizer for use under field conditions.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-200/8	Aluminum Alloy Bar, Rod, Shapes, and Tube, Extruded, 6061.
QQ-A-200/9	Aluminum Alloy Bar, Rod, Shapes, and Tube, Extruded, 6063.
QQ-A-250/8	Aluminum Alloy, 5052, Plate and Sheet.
QQ-A-601	Aluminum-Alloy Sand Castings.
QQ-P-416	Plating Cadmium (Electrodeposited).
TT-E-516	Enamel, Lustreless, Quick-Drying Styrenated Alkyd Type.
ZZ-R-765	Rubber, Silicone: Low-and High-Temperature and Tear Resistant.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055	Barrier Material, Waterproofed, Flexible.

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Military

MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys, Spot and Seam.
MIL-M-7298	Manual, Technical, Commercial Equipment.
MIL-W-8604	Welding of Aluminum Alloys, Process For
MIL-P-52192	Primer Coating, Epoxy.

STANDARDS

Federal

FED-STD-595	Colors.
FED-STD-66	Steel, Chemical Composition and Hardenability.

Military

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

DRAWINGS

Defense Personnel Support Center (DPSC)

25050	Sterilizer, Surgical Instrument and Dressing, Pressure, Direct Steam or Electrically Heated, 16 in. x 24 in.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Designation B16	Specification for Free-Cutting Brass Rod, Bar and Shapes for Use in Screw Machines.
ASTM Designation B145	Specification for Leaded Red Brass and Leaded Semi-Red Brass Sand Castings.
ASTM Designation B209	Specification for Aluminum-Alloy Sheet and Plate.
ASTM Designation B210	Specification for Aluminum Tube, Drawn, Seamless.
ASTM Designation B247	Specification for Aluminum-Alloy Die Forgings.
ASTM Designation B408	Nickel-Iron Chromium Alloy Rod and Bar.

(Copies of ASTM Standards may be obtained upon application to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA No. 70 National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts.)

U.S. PHARMACOPOEIAL CONVENTION, INCORPORATED

Pharmacopoeia of the United States

(Application for copies should be address to the Mack Publishing Company, Easton, Pennsylvania.)

3. REQUIREMENTS

3.1 Fire and casualty hazards. Switches, attachment plugs, receptacles and other components handled by the user shall be mounted securely. Terminals solely for connection of equipment-grounding conductors shall be in accordance with the National Electric Code, National Fire Protection Association (NFPA) pamphlet No. 70. Wiring terminals shall be provided with appropriate soldering lugs or pressure terminal connectors securely fastened in place and shall be prevented from turning. Minimum conductor wire size shall be No. 14 AWG for branch circuits. Power cords shall be types SJ and SJT as indicated in DPSC Drawing 25050. Type HSJ or HSJO cords shall be used where temperatures higher than 250° F. are attained on any surface that the cord can touch. Strain relief shall be provided on the power cord so that the cord shall be capable of withstanding, for one minute without displacement, a pull of 35 pounds, with the connections within the sterilizer disconnected. Other than terminal rods

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and terminal plates, unplated iron or steel shall not be used for the current-carrying parts of the sterilizer. Internal wiring shall be routed away from sharp edges, screw threads and other abrading conditions. Switches and other control devices shall have ratings not less than that of the load controlled. The current-carrying capacity of a switch which controls an inductive load, such as a transformer, shall be not less than twice the rated full-load current of the transformer.

Compliance with the above requirements regarding fire and casualty shall not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 Materials. Materials shall be as specified hereinafter. All materials employed in construction of the sterilizer assembly,including the electrical components, shall be impervious to damage from fungi. Materials welded, connected or bolted together shall be those which do not cause electrolysis. Materials shall be as indicated in Drawing 25050 and Bill of Materials thereto.

3.3 Construction. The sterilizer shall be suitable for sterilizing and drying laboratory instruments and utensils, for sterilizing flasked liquids, and for processing heat-sensitive laboratory substances by nonpressure operation. The unit shall operate safely at steam pressures equivalent to the temperatures of 250° to 254° F., and 270° to 272° F. The sterilizer shall be capable of operation from a direct steam supply, or from steam generated by the internal electrical immersion heaters.

A support structure, chamber assembly, door assembly, master controls with instrumentation, and power systems shall comprise the unit. The spare parts listed in 3.3.6 shall be furnished with the sterilizer.

The sterilizer shall be constructed in conformance with DPSC Drawing 25050 and this specification.

3.3.1 Support structure. A rigid support structure, complete with side panels, shall completely enclose the sterilizer. The top panel shall be usable as a convenient work surface. It shall be possible to install the sterilizer with the rear panel against a wall. The overall outside dimensions of the enclosure shall be 32 inches long, 20-1/4 inches wide and 36 inches high.

3.3.2 Chamber assembly. The chamber assembly shall be comprised of inner and outer shells, a backhead, a door frame, shelf rack and shelves. Inside chamber dimensions shall be not less than 16 inches in diameter by 24 inches in length. There shall be no permanent set or deflection when the sterilizer is subjected to the hydrostatic test specified in 4.4.2. The two shells shall be placed one within the other, evenly spaced and reinforced to form a jacket

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over the entire exterior of the cylindrical portion of the inner shell. The lower half of this jacket shall hold enough water, with one filling, to generate steam pressure for at least two complete sterilizing cycles, when tested as specified in 4.4.8.1. The steam generator shall have a water level gage. A valve shall be provided to drain water from the jacket. A deflector plate, at the point where the steam enters the chamber, shall be provided.

3.3.3 Door assembly. The door assembly shall consist of an opening and closing mechanism, with radial locking arms and two-stage locking mechanism. Door and mounting shall permit unobstructed access to the chamber when door is open. Door locking mechanism shall include a handwheel assembly to loosen the holding arms, but not to disengage them from the door frame channel, and a throw arm which must be activated by the operator before the holding arms can be removed from the channel. This arrangement shall allow the operator to release any residual chamber pressure gradually before disengaging the holding arms from the door frame, and thus prevent a sudden outrush of scalding steam from the chamber. The door gasket shall be silicone rubber in conformance with ZZ-R-765, Class IIb, Grade 60.

3.3.4 Manual master control and instrumentation. A single-handed multiport selector valve system shall control the operation of the sterilizer and shall provide for all the required processes of the sterilization cycle, namely, off (all ports closed), steam to chamber, slow exhaust, fast exhaust, and vacuum drying.

The following instruments shall be provided:

(a) A steam jacket pressure gauge graduated in 1-psig increments over the range of 0 to 60 psig. It shall be accurate within 1-1/2 psig over the entire range when tested as specified in 4.4.7.7.

(b) A combination vacuum-pressure gauge for the chamber. The gauge shall be graduated in 1-pound increments of pressure over the range of 0 to 60 psig and shall also show vacuums of 10, 20, and 30 inches of mercury. It shall be accurate to $\pm$ 1-1/2 psig over the entire pressure range when tested as specified in 4.4.7.8.

(c) A mechanical cycle timer adjustable for any interval between 0 and 60 minutes. It shall be graduated in 1-minute intervals and include a device that will emit an audible signal upon completion of the timed interval. When tested in accordance with 4.4.7.2, the timer shall be accurate within $\pm$ 2 seconds between 1 and 15 minutes; + 2, -0 minutes between 15 and 60 minutes.

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(d) A dial-type indicating thermometer graduated in 1-degree F. increments over the range of 50 to 300 degrees F. It shall be accurate within ± 1° F, between 210° and 270° F. when tested as specified in 4.4.7.1.

(e) The chamber shall be provided with a safety valve meeting ASME standards and set at 33 psig ± 3 psig.

3.3.5 Power system. The power system shall permit interchangeable employment of an electric power unit or direct steam from an independent source delivered at 50 to 80 psig. It shall not be necessary to remove or disconnect one power unit in order to emply the other. The electric power unit shall include immersion heaters suitable for use on 115 volts, 400 cycles, 3 phase, AC. The heating elements shall be rated at 2500 watts per element, with three elements per chamber for a total of 7500 watts. A manual-reset type low-water cutoff shall disconnect the electric power to the heaters should water in the jacket fall to unsafe level. When the cutoff has operated, is shall not revert to the "ON" position unless manually reset. The manual reset shall pass the test specified in 4.4.7.5. The pressure regulator for the electric power unit shall be accurate within ± 2 psig over its entire operating range when tested as specified in 4.4.7.4. The direct steam assembly shall have an adjustable steam pressure regulator to reduce external steam supply pressure, and to control steam pressure thus supplied to the jacket, and a thermostatic trap. The pressure regulator shall be accurate within ± 1 psig over its entire operating range when tested as specified in 4.4.7.6. The trap shall be preceded by a strainer, and connected so that condensate can be discharged from the chamber and, in addition, to permit a slight discharge of steam to maintain a constant circulation of the steam during the entire sterilization period.

3.3.6 Spare parts. The following spares shall be part of the sterilizer:

Item	Quantity	Dwg. 25050 Part No.
Gasket, Door	1	23/102
Heating Element	3	2/148
Sleeve, Compression Fitting; for Heating Element	6	5/148
Gasket, Heater Plate	1	3/148

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Item	Quantity	Part No.
Glass, Water Gage	1	82
Ring, Seal; for Water Gage	3	81
Disc, End; for Water Gage	2	80
Lamp, Neon, NE-51	1	158

3.4 Performance.

3.4.1 Temperature and pressure come-up. When tested in accordance with 4.4.8.1, each sterilizer assembly shall achieve pressure in the steam jacket and sterilizing chamber as specified in table I.

TABLE I - Steam enclosure pressure and sterilizing chamber temperature come-up times

METHOD OF OPERATION	MAXIMUM TIME (in minutes) TO ACHIEVE PRESSURE IN STEAM JACKET		MAXIMUM TIME (in minutes) TO HEAT THE STERILIZER CHAMBER	
	18 psig	30 psig	250° to 254° F.	270° to 272° F.
On steam under pressure generated by electric heaters.	22	25	27	30

3.4.2 Bacteriological function. The sterilizer assembly shall be capable of killing 100 percent of spores on test strips (see 6.5) when the assembly is tested as specified in 4.4.8.2.

3.4.3 Fast exhaust. The time required to reduce steam pressure in the sterilizing chamber from 30 psig to ambient atmospheric shall not exceed 2 minutes when tested as specified in 4.4.8.3.

3.4.4 Drying system. Immediately upon its removal from the sterilizing chamber, moisture retained by the package, shall not exceed 3 percent of the original weight of the package following the test specified in 4.4.8.4. In addition, the test package shall, following the test sterilizing cycle, be free of scorch. With sterilizer door closed, sterilizer shall be capable of drying a full load to a condition suitable for immediate use or for storage within 10 minutes (15 minutes for fabrics) following sterilization.

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3.4.5 Solutions. Exhausting the sterilizing chamber following the test specified in 4.4.8.5 shall be accomplished within 27 minutes; without blowing the bottle stoppers; with average loss of not more than 5 percent of the liquid from the load; and with normal automatic sealing of the bottles.

3.4.6 Ruggedness. The assembly shall, without physical damage or impairment of its operating efficiency, withstand the drop test specified in 4.4.9.

3.5 Finish. All metal components of the sterilizer exposed to view, except the integral housing, back of door, chamber inner walls, rack and shelves, shall receive a primer coat in conformance with MIL-P-52192 followed by not less than two coats of enamel in conformance with TT-E-516. Enamel color shall be 34554 of FED-STD-595. Aluminum components shall have finish P513; final film designation C. Steel components shall have finish P213; final film designation C. Cadmium plating shall conform to QQ-P-416 for type II. Painted components shall be finished in color X24087 of FED-STD-595.

3.6 Identification marking. Two metal nameplates shall be provided and these shall be fastened to each assembly, as specified below, by rivets or drive screws. Characters shall be white on a background painted flat black or the color specified in 3.5.

3.6.1 Government identification. This nameplate shall be 3-5/8 inches wide by 4 inches long and shall contain the following: item identification, manufacturer's order number; design pressure of the sterilizing chamber; volts; cycles and phase of heaters; Federal Stock Number; manufacturer's name; and U.S. Government contract. The plate shall be affixed to the outside of the backhead.

3.6.2 Manufacturer's identification. This plate shall be 1-1/2 inches wide by 3-1/2 inches long, legibly and durably imprinted with the following: model number, serial number, U.S. Government order number, voltage, phase, cycles, maximum kilowatts and maximum amperage. The plate shall be affixed to the outside of the backhead.

3.7 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be

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submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract, or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon requests to the Contracting Officer, Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

3.7.1 Preventive maintenance instructions. The service data shall include the following preventive maintenance instructions in order to prevent possible collapse of the sterilizer chamber:

"a. If available, use only distilled water. If this is not possible, use water as pure as practicable, demineralizing and/or filtering, as required, to assure that it is free of mud, sediment, and minerals as far as it is practical.

"b. Use care in filling the sterilizer jacket, to prevent overflow of water. When filling, vent air from the jacket by manually actuating the safety valve or by turning the selector valve knob temporarily to the "Sterilize" position, with the chamber door open. Under no circumstances should the sterilizer be hosed down for cleaning.

"c. Daily, check the low-water cutoff and water gage (sight glass) for proper functioning, by opening the drain valve and draining the jacket while the heating elements are turned on (indicated by glow of the red pilot light). Caution: The drain valve should be piped to a container, to prevent operator injury from live steam. The low-water cutoff should actuate, to turn off the power (extinguishing the pilot light), by the time, the jacket is drained. The water gage should now show empty.

"d. Daily, check the pressure control switch in accordance with the Preliminary Procedure of the sterilizer service data manual.

"e. Daily, manually actuate the "relieve handle" of each safety valve, with steam pressure in the jacket, to assure that these valves are operable.

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Utilizing the foregoing preventive maintenance procedures, the operator can readily determine the serviceability of the sterilizer. If the low-water cutoff fails to actuate or the water gage does not show empty, remains full, when checked as described above, the operator should, if practicable, discontinue use of the unit until a medical equipment repairman is available to restore it to serviceability. Under emergency conditions, an unservicable unit may be utilized, provided that the operator insures that it contains sufficient water in the jacket at all times."

3.8 Workmanship. The sterilizer assembly shall be new and free from defects and imperfections that might affect the safety, serviceability and appearance of the finished product. All parts shall be free of sharp edges and burrs. The sterilizer shall be first class, well made, and without any operating or other faults.

3.9 Welding. All welds shall be continuous. Welding of aluminum alloys shall be in accordance with MIL-W-8604. Spot welding, where used, shall be in accordance with MIL-W-6858. Welding shall be durable, sound in every detail and in accordance with latest practices recommended by the American Welding Society. Welding shall be performed only by operators experienced in performing the type of weld, and using the materials required by this specification. Prior to engaging in production welding, the welder shall demonstrate satisfactory workmanship by means of tests specified in section 4 and by conformance with the Performance Qualification paragraphs of Section IX, Welding Qualifiacations, ASME Boiler and Pressure Vessel Code.

3.9.1 Preparation of surfaces. Prior to welding, surfaces shall be free from grease, coatings or other foreign matter which could adversely effect the weld. The ends of cylinders shall be filed or machined smooth. Rough, torn, or sheared edges are not suitable.

3.9.2 Joints. All weld joints shall comply with drawing requirements respecting type, location and dimensions. Where practicable, all weld joints shall flare into the adjoined metal in gradual smooth curves, and be sealed on all sides. Beads shall be smooth and free of slag when fluxes are used. Burn-through and undercut shall not be acceptable. If fluxes are used, the excess shall be removed by a suitable method promptly after welding. There shall be no visible evidence of cracks, voids, porosity or inclusions in the weld, nor evidence of burning or overheating in the base metal. Throughout the cylinders to backhead and door ring areas, the fillet weld head contour shall be flat or slightly convex. Weld beads shall not be concave. There shall be no notches (incomplete fusing) at roots of cylinder welds. Porous welds on chambers shall be unacceptable. The proper gun angle must be maintained during the shell to backhead welding operation. Rotating the part will assist uniformity.

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3.9.3 Weld Test Specimen Failure. Failure of the Weld Test Specimen to comply with 4.4.1.2, 4.4.1.3 and 4.4.1.4 shall constitute disqualification of the welding operator for production welding and rejection of all units represented by the weld test specimen.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the sterilizer, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling for acceptance inspection.

4.2.1 For weld inspection of sterilizing chamber.

4.2.1.1 Weld Test Specimen. Prior to producing welded sterilizer chamber assemblies, each welding operator shall prepare one Weld Test Specimen for weld inspection and approval.

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Only previously approved operators shall produce welded assemblies. During production of welded assemblies, each operator shall prepare one Weld Test Specimen for each uniform lot of welded assemblies completed. The Weld Test Specimen shall be inspected and approved prior to continuing production. Non-approval of the Weld Test Specimen shall constitute Inspection Rejection of the lot of completed assemblies for either re-working or final rejection. The weld test specimen shall consist of three pieces of production material used to fabricate the sterilizing chamber:

(a) one 5" by 8" piece of flat back plate

(b) one 5" by 8" piece of inner shell material, curved to radius of inner shell

(c) one 5" by 8" piece of outer shell material, curved to radius of outer shell.

The inner and outer shell pieces shall be welded along one 8" edge to the flat surface of the piece of back plate, in the same fashion as the shells are welded to the back plate in production. When completed, it shall exactly simulate a section cut from a production sterilizing chamber unit, and shall have received the same cleaning, welding and finishing procedures as this unit.

4.2.2 Sampling for group A examination and test. The chamber assembly of each sterilizer in each lot shall be subjected to the group A examination and test specified in 4.3.1.

4.2.3 Sampling for group B examination and test. A random sample of sterilizer assemblies shall be selected in accordance with table II from each lot and shall be subjected to each of the group B examinations and tests specified in 4.3.2.

TABLE II - Sampling for group B examination and tests
AQL (approx.) = 1.5 percent defective

Number of sterilizer assemblies in lot	Number of sterilizer assemblies in sample	Acceptance number	Rejection number
		Number of sterilizer assemblies non-conforming on any group B examination and test	
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

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4.2.4 Sampling for group C tests. A random sample of sterilizer assemblies shall be selected from each lot in accordance with table III and shall be subjected to each of the group C tests specified in 4.3.3.

TABLE III - Sampling for group C tests

Number of sterilizer assemblies in lot	Number of sterilizer assemblies in sample	Acceptance number	Rejection number
		Number of sterilizer assemblies non-conforming on any group C test	
15 and under	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 and over	15	1	2

4.2.5 Sampling for group D tests. The first sterilizer assembly in the first lot shall be subjected to the group D tests specified in 4.3.4. The group D tests specified in 4.3.4. The group D tests may be waived upon request to the Contracting Officer, providing that the manufacturer can produce evidence, including certified test records, that sterilizer assemblies of the same type and design were accepted by the Government under a previous contract.

4.3 Acceptance inspection.

4.3.1 Group A examination and test. Each chamber assembly (less rack and shelves) shall be subjected to the hydrostatic test specified in 4.4.2, and to the safety and relief valve test specified in 4.4.7.3, and the results of the test compared with the requirements of this specification. Assemblies failing to pass the test shall be rejected. Those assemblies failing the test because of minor leaks at weld points may be reworked and retested.

4.3.2 Group B examination and tests. Each sterilizer assembly selected in accordance with 4.2.3 shall be subjected to each of the examinations and tests specified herein and the results of each examination and test compared with the requirements of this specification. Failure to conform to this specification for any group B examination and test shall be counted as a defect and the assembly shall be rejected. If the number of such nonconforming sterilizer assemblies in any sample exceeds the acceptance number for that sample, the lot represented by sample shall be rejected.

Group B examinations and tests	Reference paragraph
Adjustment, fit and materials	4.4.3
Dimensions	4.4.4
Wattage	4.4.5

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4.3.3 Group C test. Each of the sample sterilizer assemblies selected in accordance with 4.2.4 shall be subjected to the tests specified herein, and the results compared with the requirements of this specification. Failure to conform to this specification for the group C tests shall be counted as a defect and the assembly shall be rejected. If the number of such nonconforming sterilizer assemblies in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

Group C test	Reference paragraph
Accuracy of components	4.4.7

4.3.4 Group D tests. Each of the sample sterilizer assemblies selected in accordance with 4.2.5 shall be subjected to each of the tests specified herein, and the results of each test compared with the requirements of this specification. No sterilizer assembly shall be accepted for delivery until these tests have been satisfactorily completed. Failure of the sterilizer assembly to pass any of the tests shall result in the rejection of all units so manufactured until the cause of rejection has been corrected on all assemblies and the sample has been successfully retested. Sterilizer assemblies of the same type and design need not be retested on subsequent contract or orders. A certified record of the test results shall therefore be retained by the manufacturer.

Group D tests	Reference paragraph
Performance	4.4.8
Ruggedness	4.4.9

4.4 Examination and tests. Tests shall be conducted to determine compliance with this specification. Distilled or demineralized water only shall be used in testing the sterilizer. After testing, all residual water shall be drained and the unit thoroughly dried, using compressed air to blow water out of the tubing.

4.4.1 Apparatus and instruments. Apparatus and instruments employed for the testing shall be the laboratory precision type. They shall be checked carefully to determine that they have been installed durably and accurately. They shall be calibrated properly to assure laboratory accuracy. Performance characteristics shall be timed by a stopwatch; chamber temperatures shall be measured by the dial-type thermometer in the chamber drain line; steam jacket pressures by the steam jacket pressure gauge.

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4.4.1.1 Visual inspection. The weld test specimens shall be examined for compliance with all requirements of 3.3 and subparagraphs thereto.

4.4.1.2 Destructive Testing. Following the visual inspection of 4.4.1.1 the Weld Test Specimen shall be cut into five similar pieces, by making four transverse cuts on the 8" length. Each piece shall have a backplate measuring approximately 1-1/2" x 5" and shall be a miniature replica of the original test specimen. Both of the end pieces shall be discarded. Two of the three center test pieces shall be used for fracture inspections of the inner and outer shell welds; one set of fractures shall be produced by bending the shells away from their fillet welds, and the other set, by bending the shells toward their fillet welds. In each case the moment of bending shall be at the weld areas. If fracture does not occur at a 90° bend of the shell, the sample shall be accepted as passing the test and no further inspection shall be required. Welds which fracture easily or at bending of 90° or less shall be unacceptable.

4.4.1.3 Weld Fracture Inspection. The weld fractures shall be visually inspected under a magnification of 10 diameters. The fracture shall be free of excessive porosity, discoloration, inclusions and evidences of cracks, laminations, overlapping or undercutting. The weld shall exhibit evidences of adequate fusion with respect to root and also joint penetration.

4.4.1.4 Microscopic Inspection. The third center piece of Weld Test Specimen from 4.4.1.2 shall have one of its saw-cut surfaces prepared by grinding and polishing methods suitable for microscopic inspection of the welds as seen on cross section. Magnification shall be nominally at 10 diameters. Either Kellers or other suitable etches shall be used to reveal the fusion zone. With reference to suitable depth of weld penetration, 50% to 75% of the backplate shall be considered adequate. The microscopic inspection procedure need be followed only on the preproduction weld test sample unless failure results on the tests identified as Weld Fracture Inspection (4.4.1.3).

4.4.2 Hydrostatic test. The steam jacket, with filling funnel and plug in place, shall be tested with a hydrostatic pressure of 80 pounds per square inch (psig) for 10 minutes during which there shall be no leaking or sweating. The unit shall be examined for permanent set or deflection after the pressure has been released.

4.4.3 Adjustment, fit and materials. The sterilizer assemblies shall be subjected to a thorough examination to ascertain that the materials, workmanship and design are in strict conformance with the requirements of this specification, contract and approved drawings. The fit of parts shall be observed.

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All spare parts shall be examined to determine that they are duplicates of the parts furnished in the sterilizer assemblies. If the inspector has any reason to doubt the ready interchangeability of the repair parts with the original parts, he may require a demonstration of interchangeability. The wattage of the heating elements furnished as spare parts shall be measured in accordance with the sampling for group B examination and tests.

4.4.4 Dimensions. The sterilizer assemblies shall be measured to determine whether the chamber and overall dimensions conform to the requirements of this specification and the drawing.

4.4.5 Wattage. The wattage of the sterilizer assembly at the rated voltage shall be measured to determine conformance with 3.3.5.

4.4.6 Dielectric strength. Immediately following operation of the sterilizer assembly for one complete sterilizing cycle at 270° F. and with the main electric power switch "OFF", it shall be subjected, for one minute, to a dielectric test voltage of 1880 applied between the first terminal on the terminal block and the body of the sterilizer assembly. The test shall be repeated with the following variations: once with the main electric power switch "ON", once with the power switch "ON" and the test voltage applied between terminal "A" on the magnetic contactor and the body of the sterilizer.

4.4.7 Components.

4.4.7.1 Dial thermometer. This instrument shall be checked against a certified laboratory thermometer and accuracy recorded.

4.4.7.2 Mechanical timer. The timer shall be tested against a certified stopwatch and accuracy recorded.

4.4.7.3 Safety and relief valves. The safety and relief valves shall be connected to a steam source. The report shall include the value at which each valve released the pressure and the value at which it reseated.

4.4.7.4 Pressure control component of electric power unit. The pressure control shall be tested at intervals of 5, 10, 15, 20 and 30 psig. The accuracy shall be recorded.

4.4.7.5 Low-water cutoff. The low-water cutoff shall be tested by operating the assembly (electrically) without water to determine that the cutoff functions properly. After the unit has cut off, the jacket shall be filled with water to the proper level, the reset button actuated and the assembly operated again to ascertain that the electric steam generator still functions properly.

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4.4.7.6 Pressure regulator valve. The automatic steam pressure regulator valve (3.3.5) shall be connected to a steam source. The accuracy shall be recorded.

4.4.7.7 Pressure gage for steam enclosure. The gage for measuring pressure in the steam jacket shall be tested for accuracy against a certified laboratory standard gage and the accuracy shall be recorded.

4.4.7.8 Chamber pressure gage. The pressure gage for the sterilizing chamber shall be tested for accuracy against a certified laboratory standard gage and the accuracy recorded.

4.4.8 Performance. The performance tests shall be conducted at ambient pressure of 14.7 ± 0.1 pounds per square inch absolute (psia). Unless otherwise specified, each test shall be started with the assembly at ambient 50° to 60° F. and the steam jacket containing water (40° to 45° F.) to the "FULL" mark on the water level indicator.

4.4.8.1 Temperature and pressure come-up. Two separate tests shall be conducted as follows:

First Test

Step 1 Connect the assembly to a 115 volt, 400-cycle, three-phase power supply.

Step 2 Turn master power switch "ON" to energize the heaters.

Step 3 Close chamber door tight and turn cycle programming wheel to "OFF".

Step 4 Turn pressure control knob clockwise as far as it will go.

Step 5 Record the time required for pressure in the steam jacket to reach 18 pounds psig. Then turn the pressure control knob counterclockwise until the pilot light goes out. Pressure in the steam jacket should stabilize between 15 and 18 psig.

Step 6 When pressure in the steam jacket is stabilized between 15 and 18 psig, turn the cycle programming wheel to "STERILIZE" and record time required to reach 250° F. in the sterilizing chamber.

Step 7 Exhaust the chamber and steam jacket to zero psig, turn off heaters and drain the steam jacket.

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Second test

Step 1 Repeat Steps 1 through 4, as specified for the first test.

Step 2 Record the time required for pressure in the steam jacket to reach 30 psig. Then turn the pressure control knob counterclockwise until pilot light goes out. Pressure in the steam jacket should stabilize between 26 and 30 pounds psig.

Step 3 When pressure in the steam jacket is stabilized between 26 and 30 psig, turn the cycle programming wheel to "STERILIZE" and record time required to reach 270° F. in the sterilizing chamber.

Step 4 Exhaust the chamber and steam jacket to zero psig, turn off heaters and drain the steam jacket.

4.4.8.2 Bacteriological function. Two separate tests shall be conducted operating the assembly on electric power.

First test

Step 1 Sterilization test strips conforming to 3.4.2 shall be used.

Step 2 Place four of the spore test strips on the bottom shelf of the sterilizing chamber (two at the rear and two at the front), placing each set of two strips between two gauze sponges (4" x 4"). Hold remaining two strips for viability controls.

Step 3 Close chamber door tight and operate the sterilizer in accordance with the sterilizer manufacturer's instructions. The sterilization time shall be in accordance with the sterilization test strip manufacturer's instructions.

Step 4 With 15 to 18 psig pressure in the steam jacket turn the cycle programming wheel to "STERILIZE".

Step 5 Thirteen minutes after temperature in the chamber has reached 250° F., turn the programming wheel to "FAST EXHAUST". When pressure in the chamber is reduced to zero psig, turn the programming wheel to "OFF". Then open door, remove spore strips in a manner that will preclude contamination and conduct test for sterility as recommended in the U.S. Pharmacopoeia.

Step 6 Do not refill reservoir and continue with Second Test.

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Second test

Step 1 Repeat Steps 1, 2 and 3, as specified for the first test.

Step 2 With 27 to 30 psig in the steam jacket turn the programming wheel to "STERILIZE".

Step 3 Three minutes after temperature in the chamber has reached 270° F., turn the cycle programming wheel to the "FAST EXHAUST". When pressure in the chamber is reduced to zero psig, turn the programming wheel to "OFF". Then open door, remove spore strips in a manner that will preclude contamination and conduct test for sterility as recommended in the U.S. Pharmacopoeia.

4.4.8.3 Fast exhaust. The test shall be conducted with the assembly operating on electric power.

4.4.8.3.1 Test load. The test load shall be a tray of surgical instruments. The tray (unwrapped) shall contain 100 assorted surgical instrument such as hemostats, retractors and forceps.

4.4.8.3.2 Test procedure.

Step 1 Place test load in the sterilizing chamber.

Step 2 Operate the assembly per the manufacturer's instructions for a 270° F. sterilizing cycle. When the sterilizing chamber reaches 270° F., turn the cycle programming wheel to "FAST EXHAUST" and record the time required to reduce pressure to zero psig.

4.4.8.4 Drying. The test shall be conducted using the electric power unit.

4.4.8.4.1 Test load. The test pack shall contain the following (freshly laundered and loosely packed): 2 double-muslin wrappers, 3 surgical gowns, 12 towels, 30 gauze sponges (4 inches by 4 inches), 5 lap sponges and 1 drape sheet. The pack shall be approximately 12 inches by 12 inches by 20 inches and shall weigh 10 to 12 pounds.

4.4.8.4.2 Test procedure.

Step 1 Place the test pack on its edge in the sterilizing chamber. Close door tight. Operate the assembly (following manufacturer's instructions), exposing the pack to 250° F. for 30 minutes.

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Step 2 At completion of the 30-minute exposure period, turn cycle programming wheel to "FAST EXHAUST". When chamber pressure drops to zero psig, turn programming wheel to "OFF".

Step 3 Within two minutes after turning the programming wheel to "OFF", remove pack from chamber, weigh it at once and record the weight.

4.4.8.5 Solution exhaust system. The test shall be conducted using electric power unit.

4.4.8.5.1 Test load. The test load shall be 14 sterile fluid storage bottles, each containing 1000 cc of water (see 6.6).

4.4.8.5.2 Test procedure.

Step 1 Place test load in chamber.

Step 2 Close door tight. Operate assembly per the manufacturer's instructions to expose the load for 30 minutes at 250° F.

Step 3 At end of 30-minute exposure period, turn cycle programming wheel to "SLOW EXHAUST", turn off heaters and record time required to reduce the chamber pressure to zero psig.

Step 4 Open door only about one-half inch. Wait five minutes; then remove load.

4.4.9 Ruggedness tests. The sterilizer shall be tested with the electric power unit in operating position.

4.4.9.1 Preparations.

4.4.9.1.1 Sterilizer assembly. Drain water from the assembly; arrange the rack and shelves in their normal operating position within the sterilizing chamber; close chamber door and tighten holding arms snugly, without applying undue pressure on the gasket.

4.4.9.1.2 Test site. Select a level ground surface 6 feet square that, when tested with a cone penetrometer, will indicate an average reading of 450 psi minimum. This shall be obtained by averaging the results of ten individual readings, taken 10 inches apart, each indicating at least 250 psi.

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4.4.9.2 Test instrument. The cone penetrometer shall be for measuring soil compactness and trafficability. It shall consist of a 30-degree cone having a 1/2-square inch base area, a metal staff, a proving ring, a micrometer or indicating dial with a range of 0 to 750 psi, and a handle. As the cone is forced into the ground, the proving ring will deform in proportion to the force applied. The amount of force required to push the cone slowly into the ground in the vertical plane will be indicated in psi on the dial inside the ring. The instrument employed for the test shall be as approved by the contracting officer.

4.4.9.3 Test procedure. Place the sterilizer assembly in a rope sling in the center of the test site. Hoist the sterilizer assembly so as to place its bottom surface 51 inches above and parallel with the ground. Release the assembly from the sling by burning through the support rope, so as not to disturb horizontal alignment, allowing it to fall free the entire 51 inches. Repeat this procedure with first one side of the assembly and then with one end of the assembly 51 inches above and parallel with the ground. Following the three drop tests, the sterilizer must pass each of the tests specified in 4.4.8.1 and 4.4.8.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One sterilizer, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Spare parts. Spare parts shall be packaged in one or more boxes of appropriate size and design. Each box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent spilling of contents under normal handling. In addition, the spare parts shall be placed within the sterilizer chamber.

5.1.2.2 Service data. Service data shall be sealed within a waterproof envelope and attached to the sterilizer door.

5.1.2.3 Trays. Trays and spare parts placed within the trays shall be secured to prevent movement during transit.

5.1.3 Level C. The sterilizer shall be packaged in accordance with the contractor's commercial practice.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

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5.2.1 Level A. Each sterilizer shall be packed in an exterior container constructed in accordance with PPP-B-601, overseas type or PPP-B-621, class 2. The sterilizer shall be cushioned, blocked, braced and anchored as specified in MIL-STD-1186. All box panels, except the base, shall be lined with waterproof barrier material conforming to PPP-B-1055, class L-2 (b) or M-1. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level C. Each sterilizer shall be packed in substantial commercial (export when applicable) containers of the type and size commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Boxes containing spare parts shall be marked to identify contents and quantity contained. In addition, marking shall include the legend: "SPARE PARTS".

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Sterilization of laboratory instruments and utensils and flasked liquids, and processing heat-sensitive laboratory substances under field conditions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

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6.4 This specification covers the following item listed in the Federal Supply Catalog:

FSN	Item Identification
6530-027-5261	STERILIZER, SURGICAL INSTRUMENT AND DRESSING, Pressure, Steam or Electrically Heated, Corrosion-Resisting Metal, 16 by 24 inches, 110 Volt, 3 phase, 400 cycle, AC.

6.5 Spore strips in conformance with the requirements of 3.4.2 are commercially available as American Sterilizer Co. Spordi and Castle Co., Unispore and are stocked listed under FSN 6530-477-6720, Sterilization Test Strip Set, Bacterial Spore.

6.6 The sterile fluid storage bottles referenced in 4.4.8.5.1 are stock listed under FSN 6530-935-4136 and are commercially available from the American Sterilizer Company as their catalog number SP-1000.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6530-1165

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Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-36717 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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